Technical Notes and Engineering Decisions

Assumptions

The assignment intentionally leaves several implementation details open to interpretation. The following assumptions were made during development:

1. Every image contains exactly one valid Ground Control Point (GCP) marker.
2. The provided marker coordinates correspond to the true center of the marker.
3. Shape labels are correct whenever available.
4. Missing shape labels should not prevent coordinate regression learning.
5. Test images follow a similar distribution to the training dataset.
6. During inference, the tile with the highest classification confidence is assumed to contain the target marker.

Challenges Encountered

1. Extremely Small Target Size

The original images are approximately 4096×3000 pixels while the GCP marker occupies only a very small region of the image.

Direct full-image coordinate regression would require the model to localize a tiny object within millions of pixels.

Mitigation:

- Crop-based training
- Tile-based inference
- Local coordinate prediction inside crops

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2. Risk of Data Leakage

Multiple images belong to the same physical GCP and often share very similar viewpoints and backgrounds.

A random image-level split could result in highly correlated samples appearing in both training and validation sets, leading to artificially inflated validation performance.

Mitigation:

- GroupKFold validation
- Grouping key: `project/survey/gcp_id`

This ensures complete separation between training and validation groups.

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3. Missing Shape Labels

Four training samples contain coordinate annotations but do not contain verified shape labels.

Mitigation:

- Classification loss is ignored for those samples using `ignore_index = -100`
- Coordinate regression continues to learn from these samples

This allows utilization of all available localization annotations.

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4. Class Imbalance

The Cross class contains significantly fewer samples compared to the L-Shape class.

Mitigation:

- Macro F1 Score was used for evaluation
- A multitask learning approach was adopted to improve feature sharing between tasks

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5. High Resolution Imagery

Training directly on full-resolution images would significantly increase memory requirements and training time.

Mitigation:

- Dynamic crop generation
- EfficientNet-B3 backbone
- Mixed precision training (AMP)

Design Decisions

Why Crop-Based Training?

EDA showed that markers occupy only a tiny portion of the image.

By generating marker-containing crops, the localization task becomes significantly easier and more stable.

Why EfficientNet-B3?

EfficientNet-B3 was selected because it provides:

- Strong transfer learning performance
- Good accuracy-to-parameter ratio
- Lower memory requirements than larger CNN architectures
- Compatibility with available hardware resources

Why Multi-Task Learning?

Localization and shape classification rely on similar visual features.

Using a shared backbone:

- Reduces model complexity
- Improves feature reuse
- Enables simultaneous optimization of both tasks

Why Tile-Based Inference?

Unlike training, test images do not provide marker coordinates.

Therefore a sliding-window tile strategy was adopted to search the image for the marker location.

The tile with the highest classification confidence is selected as the final prediction.

Limitations

1. The model predicts only a single marker per image.
2. Test-set quantitative evaluation is not possible because ground-truth labels are unavailable.
3. Some localization errors remain in challenging images with small or partially visible markers.
4. The inference pipeline relies on a confidence-based tile selection heuristic.

Future Improvements

Potential future enhancements include:

- Heatmap-based localization
- Higher-resolution crops
- Cross-fold ensembling

- Confidence calibration
- Full 5-fold training and model averaging

Final Outcome

The final system successfully performs:

- Ground Control Point localization
- Marker shape classification
- Full test-set inference

A complete `predictions.json` file was generated for all 300 test images, satisfying the assignment requirements while maintaining a clean, modular, and reproducible codebase.